

Visualization Charts in Power Bi

What is Power BI?

A business intelligence (BI) tool by Microsoft

Helps in data visualization, reporting, and analytics

Connects to various data sources for insights

Key Features of Power BI

Interactive dashboards & reports

Multiple data source connectivity

AI-powered insights

Custom visualizations

- Integration with Excel, Azure, and other tools

What is Power BI?

Components of Power BI

Power BI Desktop: Report creation tool

Power BI Service: Cloud-based sharing platform

Power BI Mobile: On-the-go access to reports

Power Query: Data transformation tool

Power Pivot: Data modeling engine

Importance of Power BI

Enables data-driven decision-making

Helps in real-time reporting & monitoring

User-friendly with drag-and-drop features

- Cost-effective for businesses

Secure data handling & governance

Industries using Power BI

Finance & Banking

Healthcare

Retail & E-commerce

Manufacturing

IT & Telecom

Career Opportunities in Power BI

Power BI Developer
Data Analyst
BI Consultant
Data Scientist
Business Intelligence Engineer

Required Skills for Power BI

Data modeling & visualization
DAX (Data Analysis Expressions)
SQL & database knowledge
Power Query (ETL process)
Integration with cloud platforms

Salary & Job Trends

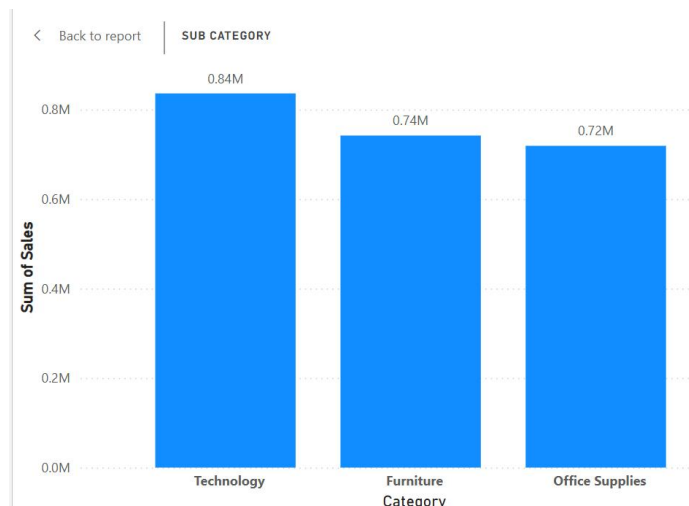
Power BI professionals are in high demand
Competitive salaries globally
Growing adoption in organizations of all sizes

Conclusion

Power BI is a powerful tool for data-driven insights
High demand in various industries
Great career opportunities with lucrative salaries

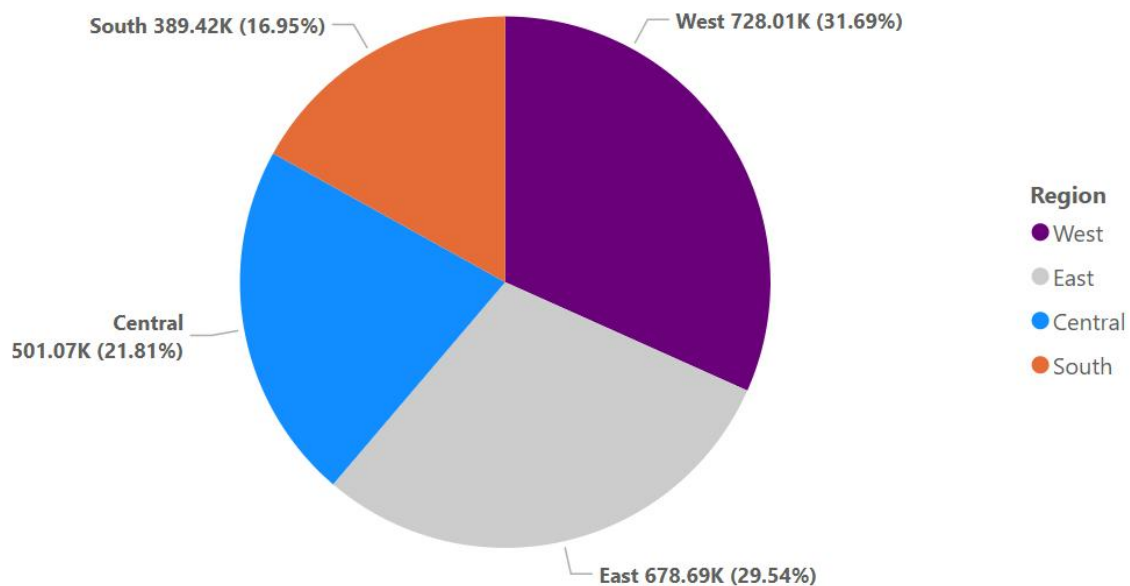
Bar Chart

A bar chart (aka bar graph, column chart) plots numeric values for levels of a categorical feature as bars. Levels are plotted on one chart axis, and values are plotted on the other axis.



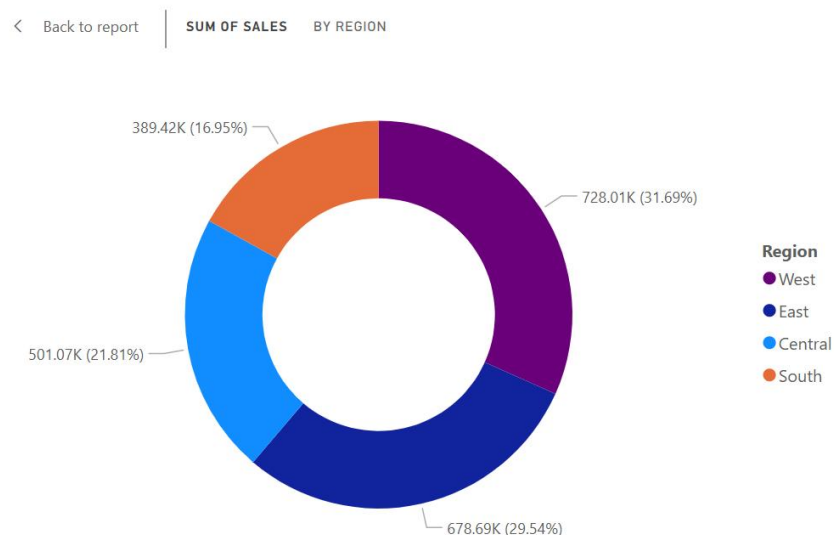
Pie Chart

In Power BI, a pie chart is a visualization tool used to represent data as a circular chart divided into slices, where each slice corresponds to a category's proportion of the whole dataset. It helps in understanding percentage distributions and comparing parts of a whole



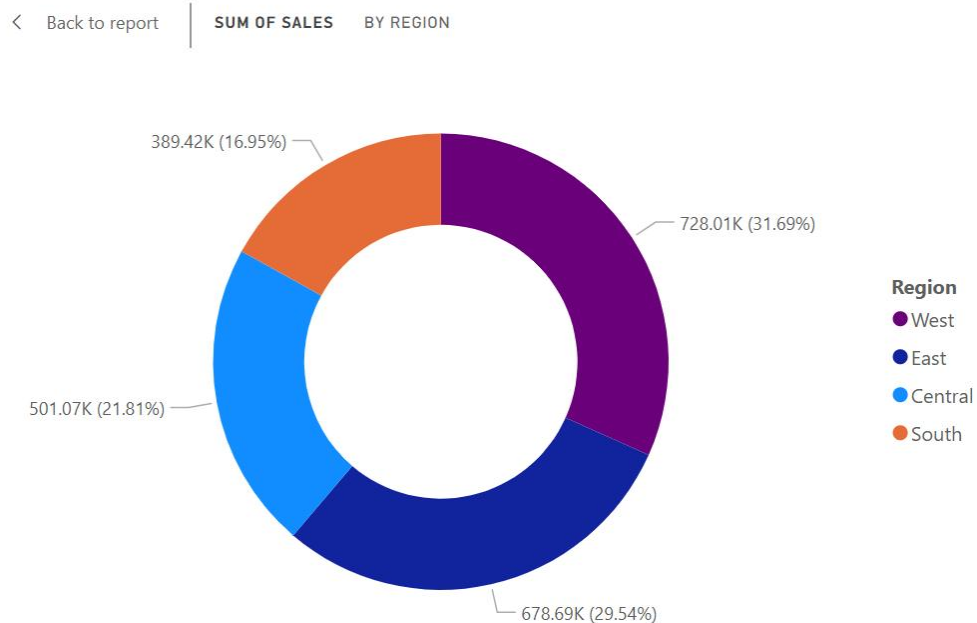
Donut Chart

A Donut Chart in Power BI is similar to a Pie Chart, but with a hollow center, making it visually distinct. It is used to represent proportions of a whole, just like a pie chart, but with additional space in the middle, which can be used for labels or KPIs



Waterfall Chart

Waterfall charts are essentially Bar charts that show a series of profits and losses for a specified time period, with a summary bar showing the amount remaining at the end of the time period. Profits appear as green bars, losses appear as red bars, and start and summary bars appear as blue bars.



Treemap Chart

TreeMap in Power BI is the hierarchical chart, which is used to show the parent and child data distribution. TreeMap is shown by a group of rectangles, these rectangles are segregated on the basis of the category. The larger numeric values are present at the top, and lower numeric values are present at the bottom.



Table and matrix

Table visualization is just a normal grid containing related data in rows and columns. It can have headers and only rows. Users can create tables in reports and cross highlight elements within the table. They can also select individual cells, rows, and columns.

Sub-Category	Sum of Sales	Sum of Profit
Accessories	1,67,380.32	41,936.64
Appliances	1,07,348.45	18,227.39
Art	27,111.51	6,525.82
Binders	2,02,986.25	30,083.39
Bookcases	1,14,880.00	-3,472.56
Chairs	3,27,717.16	26,370.58
Copiers	1,49,528.03	55,617.82
Envelopes	16,476.40	6,964.18
Fasteners	8,957.43	1,110.40
Furnishings	91,656.30	13,044.97
Labels	12,471.69	5,539.38
Machines	1,89,238.63	3,384.76

Matrix

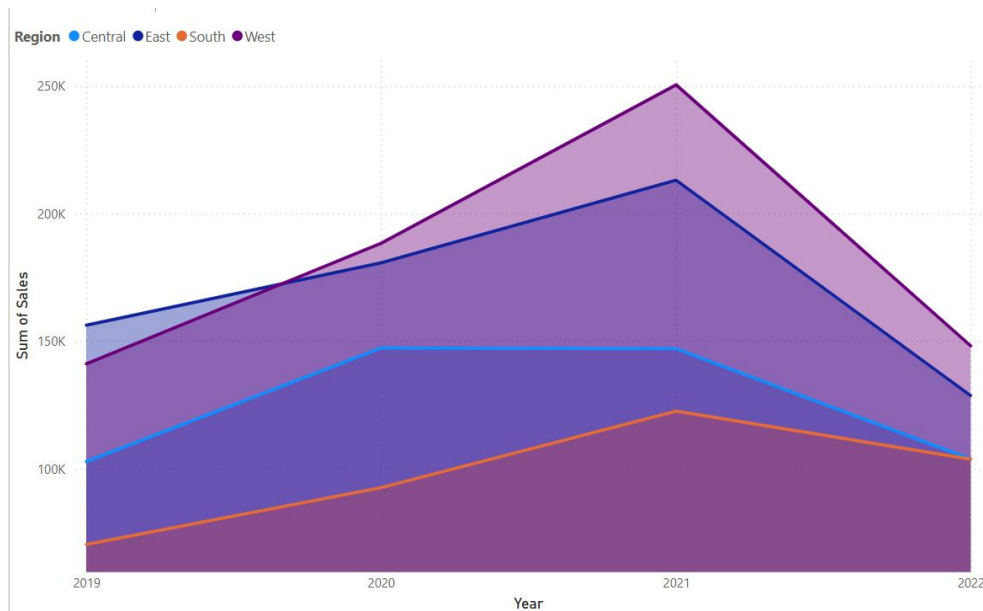
Matrix visualization is very similar to table visualization. In table visualization, data is present in a two-dimensional format. But Matrix visualization can display data in multiple dimensions

The Matrix aggregates the data itself and lets users drill down.

Region	Central		East		South	
Sub-Category	Sum of Sales	Sum of Profit	Sum of Sales	Sum of Profit	Sum of Sales	Sum of Profit
Accessories	33,956.08	7,251.63	45,033.37	11,195.86	27,196.76	6,525.82
Appliances	23,513.22	-2,514.76	34,188.47	8,391.41	19,525.33	4,655.62
Art	5,765.34	1,195.16	7,485.76	1,899.94	4,655.62	1,110.40
Binders	56,923.28	-1,043.64	53,498.00	11,267.93	37,030.34	8,391.41
Bookcases	24,157.18	-1,997.90	43,819.33	-1,167.63	10,637.40	-3,472.56
Chairs	85,230.65	6,592.72	96,260.68	9,357.77	44,444.51	26,370.58
Copiers	37,259.57	15,608.84	53,219.46	17,022.84	9,299.76	55,617.82
Envelopes	4,636.87	1,777.53	4,375.87	1,812.41	3,345.56	6,964.18
Fasteners	778.03	236.62	819.72	263.99	503.32	1,110.40
Furnishings	15,254.37	-3,906.22	29,071.38	5,881.41	17,281.44	13,044.97
Labels	2,451.47	1,073.08	2,571.43	1,114.16	2,353.18	5,539.38
Machines	26,797.38	-1,486.07	66,106.17	6,928.64	53,890.96	3,384.76
Paper	17,491.90	6,971.90	20,172.60	9,015.37	14,135.43	
Phones	72,305.30	12,295.59	1,00,614.98	12,314.69	58,098.34	
Storage	45,930.11	1,969.84	71,612.58	8,389.37	35,745.69	
Supplies	9,467.37	-661.89	10,704.52	-1,171.26	8,318.93	
Tables	39,154.97	-3,559.65	39,139.81	-11,025.38	42,958.61	
Total	5,01,073.10	39,802.79	6,78,694.14	91,491.54	3,89,421.17	

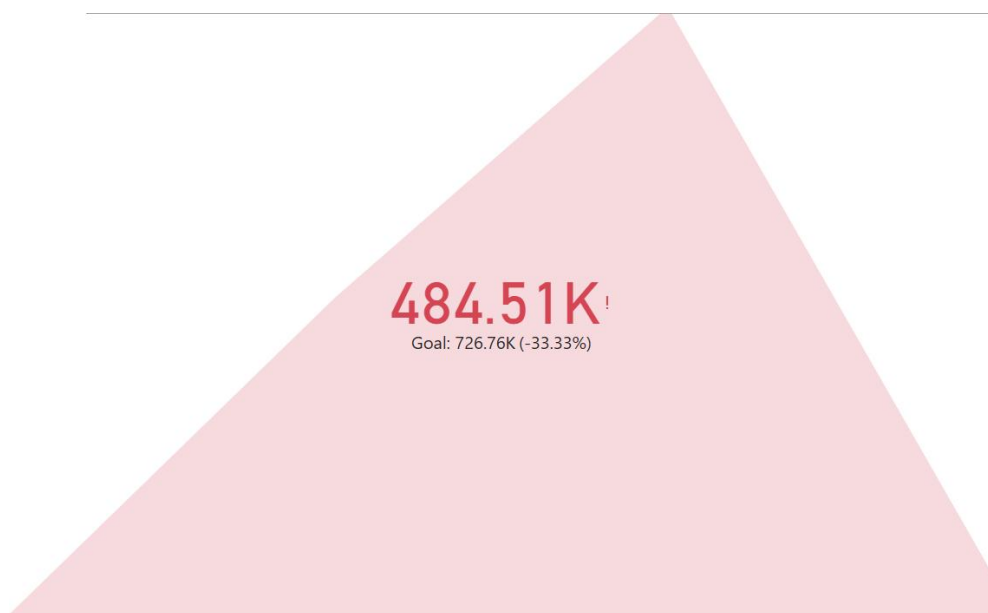
Area Chart

The basic area chart (also known as layered area chart) is based on the line chart. The area between axis and line is filled with colors to indicate volume.



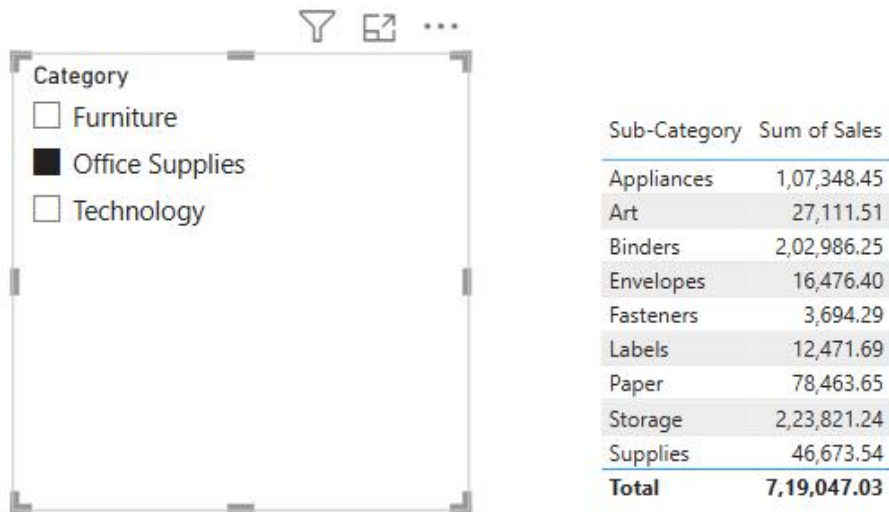
KPI Chart

The KPIs evaluate an association's performance in different areas by assessing quantifiable measure values. In businesses, KPIs play an essential part as visual cues that indicate growth by assessing values and measurable pretensions. A KPI always has a base value or measure which is estimated against a target value. That is, comparing the factual performance with the target set. You can estimate the performances in an analysis just by seeing the KPIs.



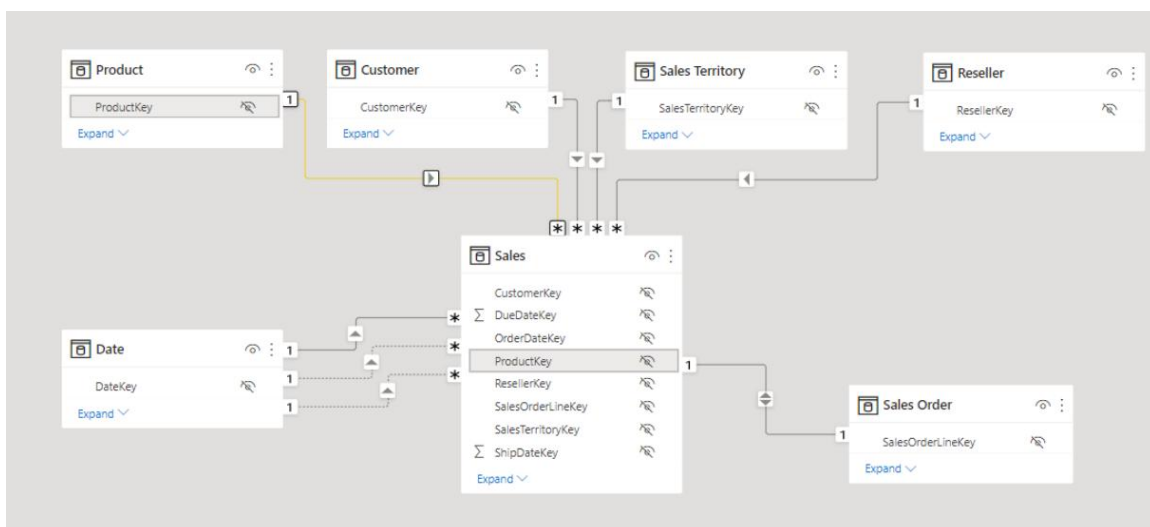
Slicers

Slicers are the interactive filters used to format and analyze the parts of the data more effectively. We can create a slicer easily with 4 steps, as shown in the later stage of the article. We have various options to format slicers, we can change the value of the alt text, title, etc. We can also add various options like slicer headings, hierarchy distribution, and multi-select slicers.



Data model

A data model in Power BI is a logical representation of data structure and relationships. It's a collection of tables and relationships between them.



Importance of Data Model

Defining Tables and Columns:

Identifying the relevant data elements and structuring them into tables with appropriate columns.

Establishing Relationships:

Creating connections between tables based on shared key values to enable data analysis across different datasets.

Data Types:

Assigning appropriate data types to columns for accurate calculations and visualizations.

Measures and Calculations:

Defining custom metrics and calculations based on the data model to answer specific business questions.

Importance of Data Model

Key benefits of a well-designed data model in Power BI:

Improved Accuracy:

By establishing clear relationships between data points, a data model minimizes data redundancy and errors, leading to more accurate reports and insights.

Enhanced Performance:

A properly structured data model optimizes query performance by reducing unnecessary calculations and data retrieval, allowing for faster analysis.

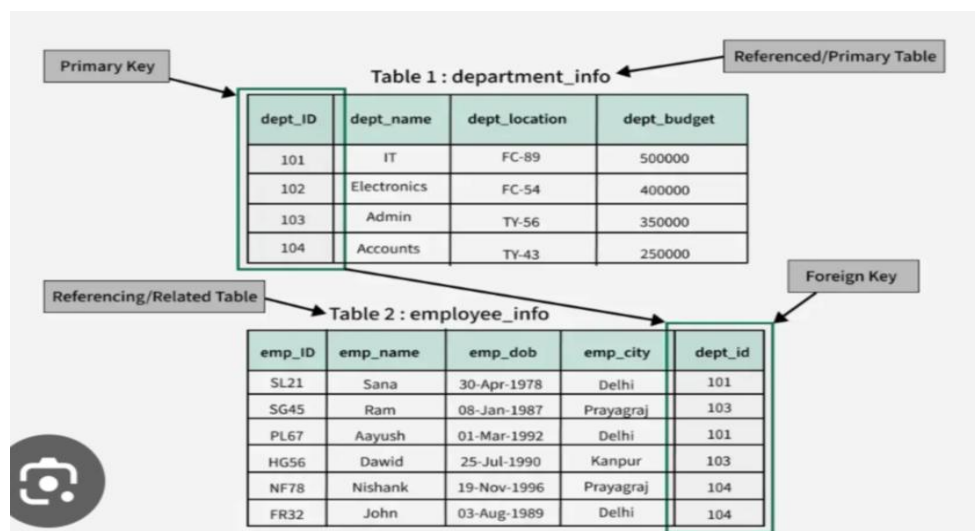
Clarity and Understanding:

A visual representation of the data model helps users easily grasp the structure of the data and how different elements are connected, facilitating better interpretation of results.

Primary and foreign key

A column or set of columns that uniquely identifies each row in a table

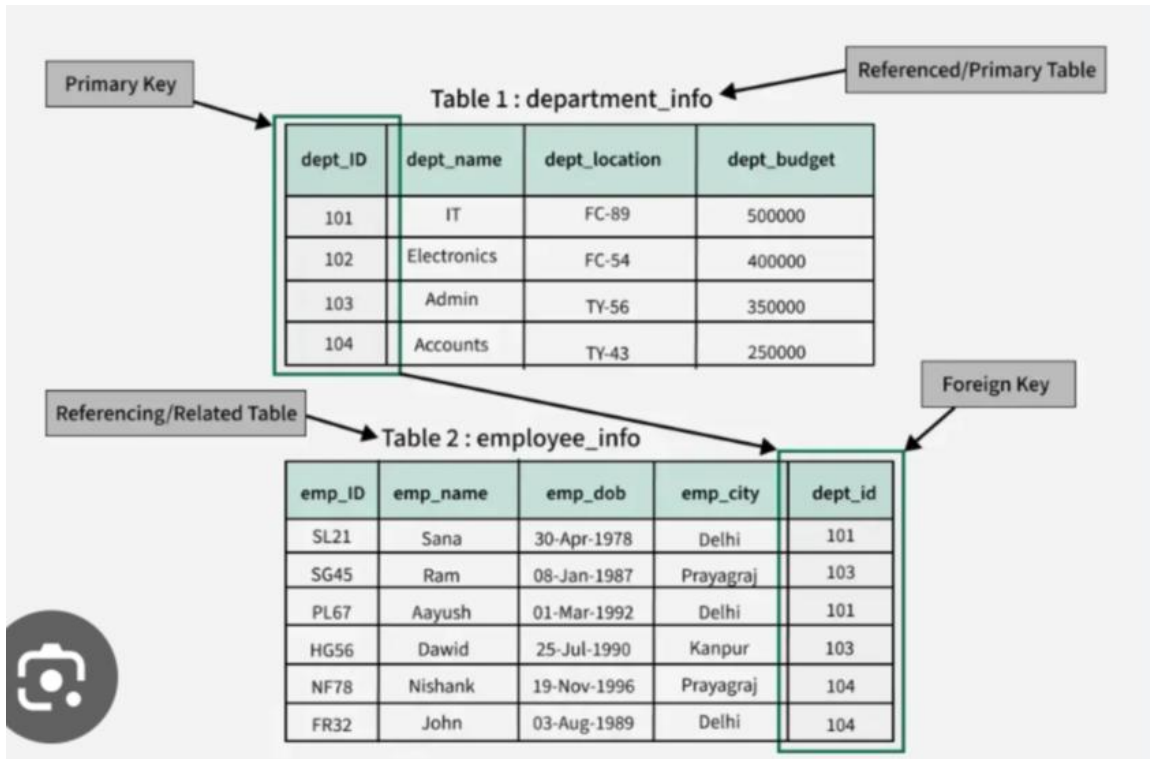
A column or set of columns that references the primary key of another table



Primary and foreign key

A column or set of columns that uniquely identifies each row in a table

A column or set of columns that references the primary key of another table



One to One

In One-to-One relationship, one record of the first table will be linked to zero or one record of another table

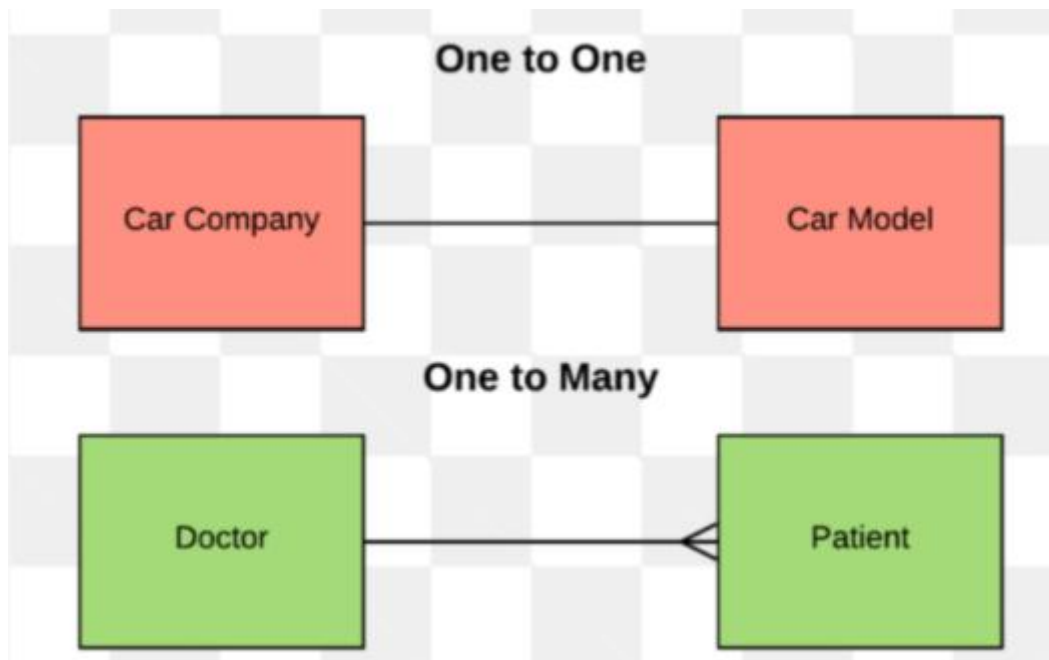
ID	Name	City	Mobile	Aadhaar No.	State	Issue Date
1	Rahul	Patna	2569874	5987458	Bihar	12-12-2000
2	Sonam	Kolkata	4587985	5981254	West Bengal	11-09-2003
3	Kunal	Mumbai	4789854	5412635	MH	07-01-2001

Person

ID	Name	City	Mobile
1	Rahul	Patna	2569874
2	Sonam	Kolkata	4587985
3	Kunal	Mumbai	4789854

Aadhaar

ID	Aadhaar No.	State	Issue Date	RID
1	5987458	Bihar	12-12-2000	1
2	5981254	West Bengal	11-09-2003	2
3	5412635	MH	07-01-2001	3



One to Many

One-to-Many is the most commonly used relationship among tables. A single record from one table can be linked to zero or more rows in another table.

Customer

CustID	Name	City	Mobile
1	Rahul	Patna	2569874
2	Sonam	Kolkata	4587985
3	Kunal	Mumbai	4789854

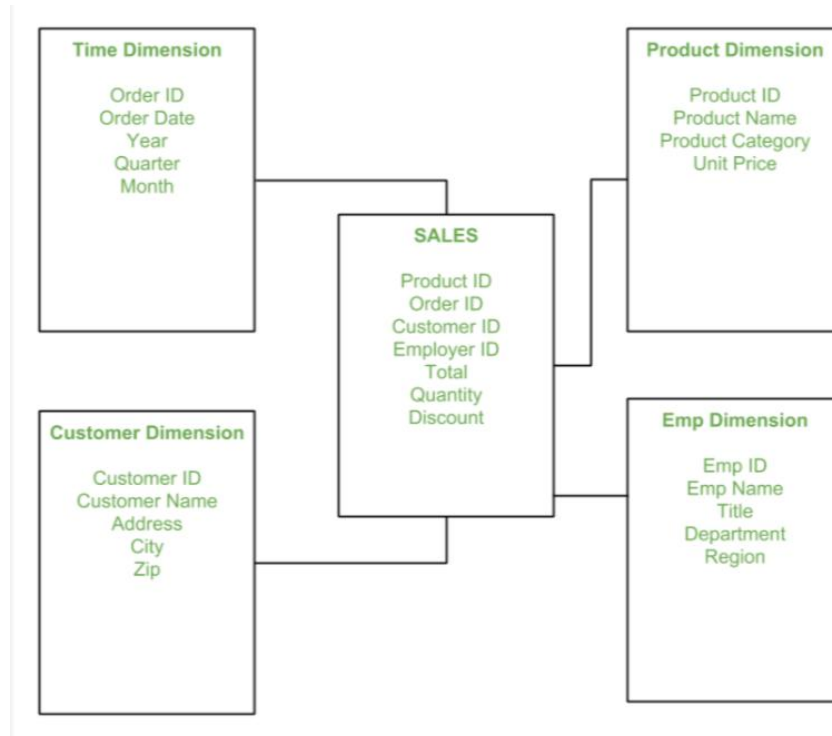
Order

OrID	Quantity	Date	CustID
451	12	12-12-2000	1
298	54	11-09-2003	1
569	17	07-01-2001	3

Data modeling-Schema

Star schema

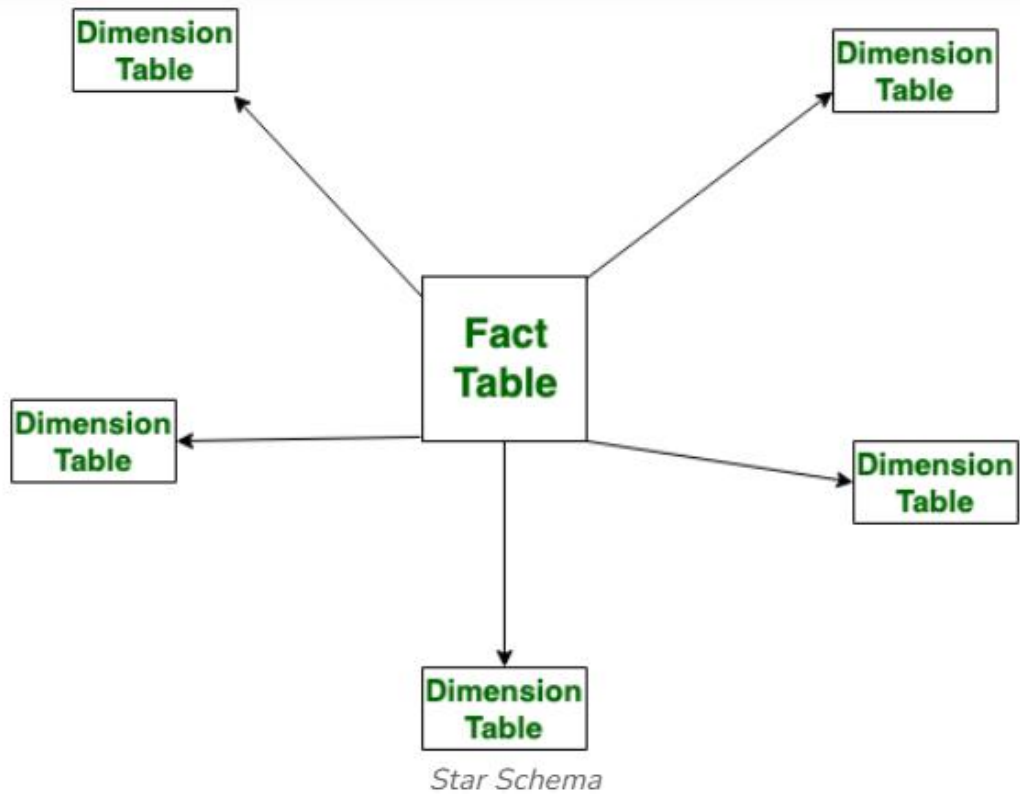
A star schema is made up of a central fact table with multiple dimension tables branching off of this fact table, much like the appearance of a star.



Fact and Dimension

The fact table in a star schema contains the measures or metrics that are of interest to the user or organization.

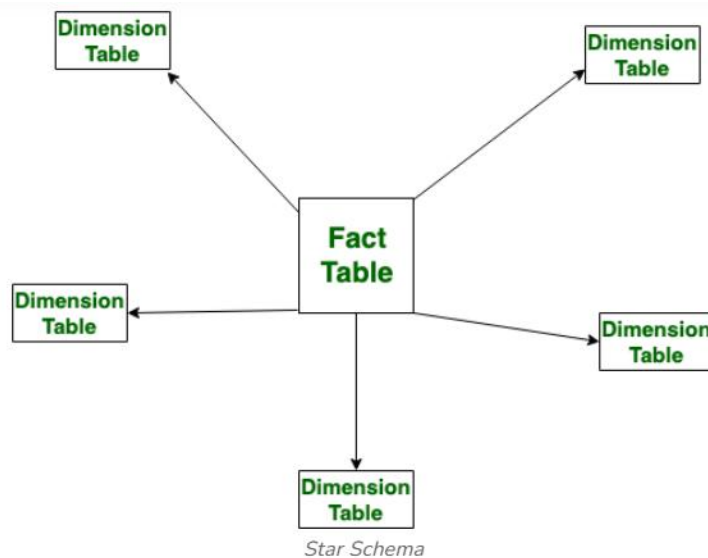
The dimension tables in a star schema contain the descriptive attributes of the measures in the fact table.



Fact and Dimension

The fact table in a star schema contains the measures or metrics that are of interest to the user or organization.

The dimension tables in a star schema contain the descriptive attributes of the measures in the fact table.



Creating and managing relationship

Creating and Managing table relationships typically refers to the process of defining and maintaining the connections between tables in a relational database. Table relationships are crucial for maintaining data integrity, enforcing referential integrity, and ensuring efficient data retrieval.

Power BI is a powerful tool to understand and then manage and edit the table accordingly. It is used in data visualization which helps in drawing business intelligence reports from data.

Autodetect Tool for creating relationships

If after loading all the tables, still no relationship can be seen between tables, then the Autodetect tool comes in handy. To use Autodetect, on the Modeling tab select Manage Relationships > Autodetect.

Manually Creating Relationships

But sometimes Power BI doesn't create the correct relationships so there's also an option of creating relationships between table manually

DAX and its importance

What is DAX?

DAX stands for Data Analysis Expressions.

It is a formula language used in Power BI, Excel, and Analysis Services.

Helps in creating calculated columns, measures, and custom tables.

Why DAX is Important?

Enhances data modeling capabilities.

Allows for efficient calculations and aggregations.

Provides flexibility in analyzing data.

Enables advanced business intelligence solutions.

DAX and its importance

Key Features of DAX

Ability to create custom calculations.

Powerful aggregation functions.

Time intelligence functions.

Relationship and filtering capabilities.

Basic DAX Functions

Aggregation Functions (SUM, AVERAGE, MIN, MAX, COUNT)

Logical Functions (IF, SWITCH, AND, OR)

Text Functions (CONCATENATE, LEFT, RIGHT, MID)

Date & Time Functions (TODAY, NOW, YEAR, MONTH)

Filter Functions (FILTER, ALL, RELATED)

DAX and its importance

DAX in Power BI

Used for creating measures and calculated columns.

Helps in dynamic reporting and data visualization.

Enhances interactivity in reports.

Example: Use Cases of DAX

Calculating total sales revenue.

Creating year-over-year growth metrics.

Implementing rolling averages.

Filtering data dynamically based on user selection.

DAX and its importance

Advantages of Using DAX

Improves report performance and efficiency.

Simplifies complex calculations.

Enhances decision-making with advanced analytics.

Challenges and Best Practices

Challenges:

Learning curve for complex formulas.

Performance issues with large datasets.

Best Practices:

Optimize DAX queries for efficiency.

Use variables for better readability.

Minimize calculated columns where possible.

Introduction to Evaluation Contexts

Power BI uses DAX (Data Analysis Expressions) for calculations.

Understanding Evaluation Contexts is key to writing efficient DAX formulas.

Two main types:

Row Context
Filter Context

Row context

Definition: Row context applies when a calculation is performed row-by-row in a table.

Created automatically in:

Calculated Columns

Iterating Iterating Functions (e.g., SUMX, FILTER)

Example:

DAX

```
TotalSales = Sales[Quantity] * Sales[Unit Price]
```

This formula operates at the row level in a calculated column.

Demonstrating Row Context

Image: A Power BI table showing calculated columns.

Key Points:

Each row is evaluated separately.

Cannot automatically access values from other rows.

Requires iterators (SUMX, AVERAGEX, etc.) to work with multiple rows.

Filter Context?

Definition: Filter context is applied when filters affect a calculation.

Created by:

Slicers, Filters, and Rows in a Pivot Table

CALCULATE Function

Here, CALCULATE modifies the filter context to show only sales from the West region.

Example:

DAX

Copy

```
TotalSalesFiltered = CALCULATE(SUM(Sales[Sales Amount]), Sales[Region] = "West")
```

Demonstrating Filter Context

Image: A Power BI report with slicers & filters.

Key Points:

Filters from visuals, slicers, or DAX affect calculations.
CALCULATE can modify or override filters.
Multiple filters can combine to create complex contexts.
Row Context vs Filter Context

Aspect	Row Context	Filter Context
Definition	Evaluates row-by-row in a table	Applies filters to calculations
Created By	Calculated Columns, Iterators	Filters, Slicers, CALCULATE
Scope	One row at a time	Entire dataset based on filters
Example Function	SUMX , FILTER	CALCULATE , ALL , KEEPFILTERS

Calculated Columns

Definition: A calculated column is a new column created in a table using DAX.
It is calculated row-by-row at the data model level.

Example:

Pros: Can be used in relationships, filters, and aggregations.

Cons: Consumes storage and affects performance for large datasets.

```
Sales[TotalPrice] = Sales[Quantity] * Sales[UnitPrice]
```

Definition: A measure is a dynamic aggregation calculated at runtime based on user interactions.

Example:

Pros: Optimized for performance, does not increase storage.

Cons: Cannot be used in calculated columns.

Two types of measure:

Implicit measure

Explicit measure

Calculated Measures

```
Total Sales = SUM(Sales[TotalPrice])
```

Purpose: Aggregates a single column by adding all values.

Syntax:

Use Case: Total revenue calculation

DAX - SUMX Function

Purpose: Iterates over a table and performs row-wise calculations before aggregation.

Syntax:

Use Case: Useful when multiplication is needed before summation.

DAX- Sum and sumX

```
Revenue = SUMX(Sales, Sales[Quantity] * Sales[UnitPrice])
```

```
Total Sales = SUM(Sales[Revenue])
```

DAX- Min,Max,Minx and MaxX

MIN: Returns the smallest value in a column.

Min Price = MIN(Sales[UnitPrice])

MAX: Returns the largest value in a column.

Max Price = MAX(Sales[UnitPrice])

Use Case: Finding the lowest or highest sales price

MINX: Returns the minimum value after performing row-wise calculations.

Min Revenue = MINX(Sales, Sales[Quantity] * Sales[UnitPrice])

MAXX: Returns the maximum value after performing row-wise calculations.

Max Revenue = MAXX(Sales, Sales[Quantity] * Sales[UnitPrice])

Use Case: Finding the lowest or highest revenue per transaction

DAX- Average, AverageX

AVERAGE Function

Definition:

Returns the arithmetic mean (average) of a column.

Syntax:

AVERAGE(column)

Example:

AVERAGE(Sales[Revenue])

This returns the average revenue.

AVERAGEX Function

Definition:

Computes the average of an expression evaluated over a table.

Syntax:

AVERAGEX(table, expression)

Example:

AVERAGEX(Sales, Sales[price] * Sales[Quantity])

DAX- Count, DistinctCount

Definition: Count

Counts the number of non-blank values in a column.

Syntax:

COUNT(column)

Example:

COUNT(Sales[OrderID])

This counts the total number of orders.

Definition: DistinctCount

Counts the number of unique values in a column.

Syntax:

DISTINCTCOUNT(column)

Example:

DISTINCTCOUNT(Sales[CustomerID])

This counts the number of unique customers who made purchases.

DAX- Date Time function

Power BI provides a variety of Date and Time functions to handle, manipulate, and analyze time-based data.

Essential for time intelligence, trend analysis, and reporting.

Functions available in DAX (Data Analysis Expressions) and Power Query.

Types of Date/Time Functions

Date Functions (e.g., TODAY, DATE, YEAR, MONTH)

Time Functions (e.g., HOUR, MINUTE, SECOND, NOW)

Date & Time Arithmetic (e.g., DATEDIFF, EDATE, EOMONTH)

TODAY() – Returns the current date.

DATE(year, month, day) – Creates a date from year, month, and day.

YEAR(date), MONTH(date), DAY(date) – Extracts respective components.

Example: YEAR(TODAY()) returns the current year.

DATEDIFF(start_date, end_date, unit) – Difference in days, months, or years.

DAX- Text/String function

CONCATENATE()

Concatenate function is used to join two columns or to create a text String by joining two text strings. The CONCATENATE function combines two text strings into a single text string.

Syntax: CONCATENATE (<Col 1> , <Col 2>)

LEFT ()

Left function returns the requested number of text characters from the left side of the given string

Syntax: LEFT (<Text>, <Num_Chars>)

RIGHT()

Depending on how many characters you enter, this function returns the last characters in the last characters in text string.

Syntax: RIGHT(<Text>, [<Num_Times>])

DAX- Text/String function

MID()

MID gives a beginning position, length, and a string of characters from the middle of a text string.

Syntax: MID(<Text>, <Start_Num>, <Num_Chars>)

UPPER()

With the help of the upper function, a text string is changed to all uppercase letters.

Syntax: UPPER(<Text>)

LOWER()

With the help of the lower function, a text string is changed to all lowercase letters.

Syntax: LOWER(<Text>)

DAX- Logical function

DAX The Logical function of Power BI is among its most crucial components. It offers up a result as either True or False.

IF

A condition is checked, and if it is TRUE, one value is returned; otherwise, a different value is returned.

Syntax: IF(<logical_test>, <value_if_true>[, <value_if_false>])

Example: Categorizing sales performance= IF(Sales[TotalSales] > 1000, "High", "Low")

This formula labels total sales as "High" if they exceed 1000; otherwise, it labels them as "Low."

AND Function:

Syntax: AND(<Logical1>, <Logical2>, ...)

Description: Returns TRUE if all arguments are TRUE

Example: = AND(Sales[TotalSales] > 1000, Sales[Region] = "North")

OR Function:

Syntax: OR(<Logical1>, <Logical2>, ...)

Description: Returns TRUE if any argument is TRUE

Example: = OR(Sales[TotalSales] > 1000, Sales[Region] = "North")

DAX- Logical function

NOT Function

Syntax: NOT(<Logical>)

Description: Reverses the Boolean value of the argument

Example: = NOT(Sales[TotalSales] > 1000)

This returns TRUE if TotalSales is not greater than 1000.

SWITCH Function

Syntax: SWITCH(<Expression>, <Value1>, <Result1>, <Value2>, <Result2>, ..., [<Else>])

Description: Evaluates an expression against a list of values and returns the result corresponding to the first matching value; if no match is found, returns the default value if provided

DAX- Logical function

Column 2 = SWITCH(TRUE(),Orders[Quantity]>=12,"excelent",Orders[Quantity]>=9,"very good",Orders[Quantity]>=6,"Good", "poor")

Example: Assigning categories based on ratings

- graphql Copy Edit

```
= SWITCH(  
  TRUE(),  
  Ratings[Score] >= 90, "Excellent",  
  Ratings[Score] >= 75, "Good",  
  Ratings[Score] >= 60, "Average",  
  "Poor"  
)
```
- This formula categorizes scores into "Excellent," "Good," "Average," or "Poor" based on the value of `Ratings[Score]`.

DAX- Logical function

Example: Creating a calculated column that assigns a discount rate based on sales volume and region

```

= IF(
    AND(Sales[TotalSales] > 500, Sales[Region] = "West"),
    0.1,
    0.05
)

```

This assigns a 10% discount if sales are over 500 and the region is "West"; otherwise, a 5% discount is applied.

Connectivity

Power BI is a powerful business intelligence tool.

It connects to various data sources for analysis and reporting.

Understanding data connectivity is key to efficient data management.

Types of Data Sources in Power BI

Files (Excel, CSV, JSON, XML)

Databases (SQL Server, MySQL, PostgreSQL, Oracle)

Online Services (SharePoint, Google Analytics, Azure)

Big Data Sources (Azure Data Lake, Hadoop, Spark)

Other Sources (APIs, Web, OData Feed, SAP)

Connection Modes in Power BI

1. Import Mode

Loads data into Power BI

Best for performance but large datasets may slow it down

2. Live Connection

Connects directly to a live dataset (e.g., SSAS)

Best for enterprise-scale models

Connectivity

Connecting to Different Data Sources

1. Open Power BI Desktop

2. Click Get Data

3. Choose a data source (Excel, SQL, Web, etc.)

4. Configure connection settings

Load data into Power

Data Transformation with Power Query

- Cleaning & Shaping Data: Removing duplicates, handling nulls
- Merging & Appending Tables

- Creating Calculated Columns & Measures
- Applying Business Rules

Creating Reports

Step 1: Get Your Data

Open Power BI Desktop.

Click Home → Get Data and choose a source (Excel, SQL Server, CSV, API, etc.).

Load the data and check it in Power Query (transform if needed).

Step 2: Model Your Data

Go to Model View to check relationships between tables.

Create relationships if necessary using Manage Relationships.

Add calculated columns and measures using DAX (Data Analysis Expressions).

Step 3: Create Visuals

Open the Report View.

Use visualizations like bar charts, pie charts, line graphs, cards, maps, etc..

Drag and drop fields into visualizations.

Use filters, slicers, and drill-through options for interactivity.

Creating Reports

Step 4: Format and Customize

Change colors, labels, and fonts using the Format Pane.

Use Bookmarks and Buttons for navigation.

Apply conditional formatting for better insights.

Step 5: Publish and Share

Save the report and click Publish to upload it to Power BI Service.

Create a Dashboard in Power BI Service by pinning key visuals.

Share the dashboard with stakeholders by giving access.

Calculate function

CALCULATE function in two parts; the aggregation and the filter. The first part of the expression is the aggregation piece. This is where you can place whatever aggregation function you'd like whether it's SUM(Sales), AVG(Price), or something else.

The second portion is your filter criteria. This part will define the data set that you are applying the aggregation in the first part to.

```
CALCULATE(<[expression]>,[filter1],[filter2],[filter...]))
```

ALL, ALL Selected, ALL Except

ALL

Returns all the rows in a table, or all the values in a column, ignoring any filters that might have been applied

ALL Selected

Returns all the rows in a table, or all the values in a column, ignoring any filters that might have been applied inside the query, but keeping filters that come from outside

ALLSELECTED([<tableName> | <columnName>[, <columnName>[, <columnName>[,...]]]]))

ALL Except

Returns all the rows in a table except for those rows that are affected by the specified column filters.

```
ALLEXCEPT(<table>,<column>[,<column>[,...]])
```

```
ALL( [<table> | <column>[, <column>[, <column>[,...]]]]) )
```

ALL, ALL Selected, ALL Except

ALL

All = CALCULATE(SUM(Orders[Quantity]),ALL(Orders))

ALL Selected

Allselct = CALCULATE(SUM(Orders[Quantity]),ALLSELECTED(Orders[State]))

ALL Except

Allexcept = CALCULATE(SUM(Orders[Quantity]),ALLEXCEPT(Orders,Orders[State]))

Cross Filter Function

Specifies the cross-filtering direction to be used in a calculation for a relationship that exists between two columns.It is used in calculate function

Crosfiltred =

CALCULATE(DISTINCTCOUNT(customers[customerID]),CROSSFILTER(orders[customerID],customers[customerID],Both))

DAX

```
CROSSFILTER(<columnName1>, <columnName2>, <direction>)
```